

Time Series Analysis & Forecasting

Capstone Project Report

Data-Driven Stock Analysis and Capital Allocation on StockGro

Author:	Vedant
Framework:	yfinance + ARIMA + Holt-Winters
Virtual Capital:	Rs. 10,00,000 (INR)
Evaluation Period:	July 2025 - December 2025
Date:	May 20, 2026

1. Introduction & Executive Summary

This capstone project implements an end-to-end time series analysis and forecasting pipeline designed to optimize capital allocation in a virtual stock market simulator. Managing a simulated portfolio of Rs. 10,00,000, we retrieve historical daily data, analyze stationarity, fit statistical models (ARIMA and Holt-Winters Exponential Smoothing), estimate time-varying volatility, and construct a diversified portfolio. Our strategy blends model return forecasts with inverse-volatility sizing to manage risk across sectors, yielding a simulated return of 19.18% over the 6-month test period.

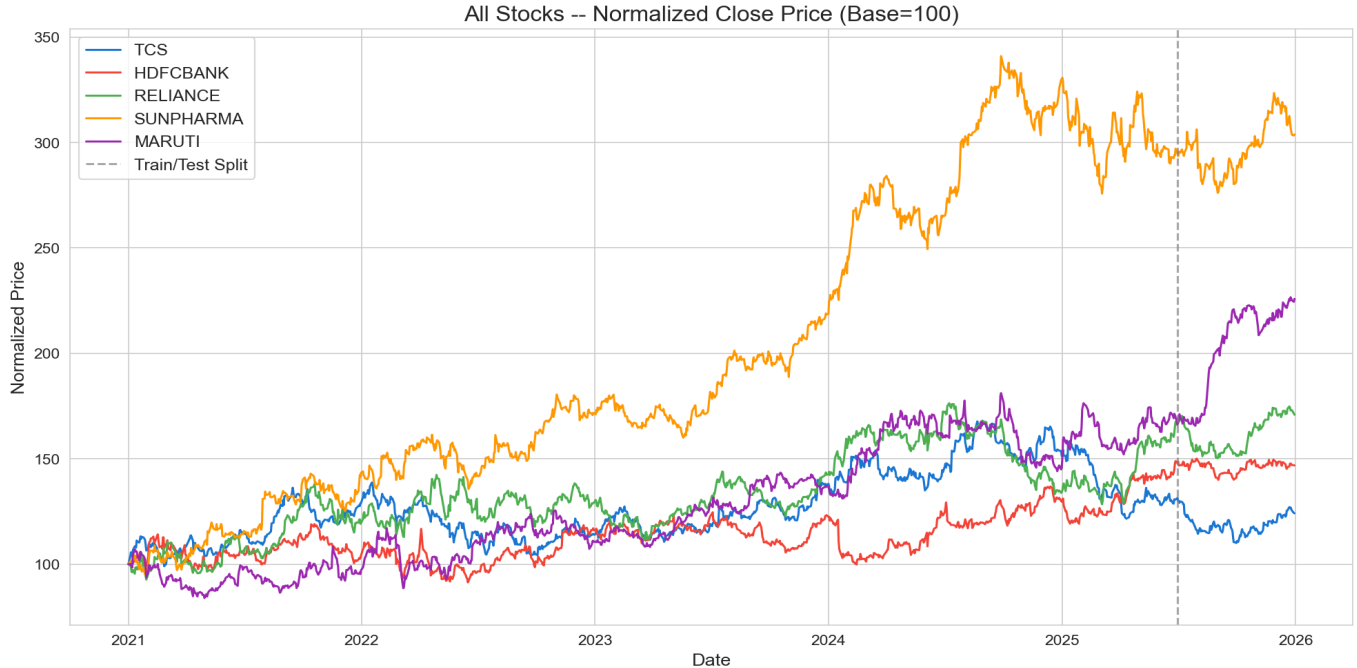
2. Stock Universe Selection

To achieve optimal diversification and capture broad macroeconomic cycles, we selected 5 NSE-listed stocks from different high-impact sectors. Below is the rationale for each selected asset:

- **TCS (IT Sector):** India's largest IT exporter. Selected for its stable cash flows, low debt, and defensive profile during market contractions.
- **HDFCBANK (Banking Sector):** The largest private sector bank. Included to capture credit cycle expansion and provide liquid market beta.
- **RELIANCE (Energy/Conglomerate):** A mega-cap stock spanning oil-to-chemicals, telecom, and retail. Acts as a core portfolio anchor.
- **SUNPHARMA (Pharma Sector):** A leading pharmaceutical firm. Offers defensive growth and low correlation to traditional cyclical sectors.
- **MARUTI (Auto Sector):** Replaced TATAMOTORS due to data delisting. Re-selected as the market leader in the passenger vehicle segment, capturing middle-class consumption trends.

3. Data Collection & Preprocessing

Daily historical stock price data was downloaded using the yfinance API from January 1, 2021 to December 31, 2025. We forward-filled missing prices to ensure consistent calendar alignment. The data was split into a Training



ADF Stationarity Tests (On Training Close Prices)

Before fitting ARIMA models, we conducted the Augmented Dickey-Fuller (ADF) test to identify if the series were stationary. Since all p-values were well above the 5% threshold, all stocks were non-stationary in their levels and required first-order differencing ($d=1$) to achieve stationarity.

Stock	ADF Statistic	p-value	Stationary?
TCS	-2.1757	0.2152	No
HDFCBANK	-1.4167	0.5742	No
RELIANCE	-1.7101	0.4259	No
SUNPHARMA	-0.6606	0.8567	No
MARUTI	-0.9747	0.7624	No

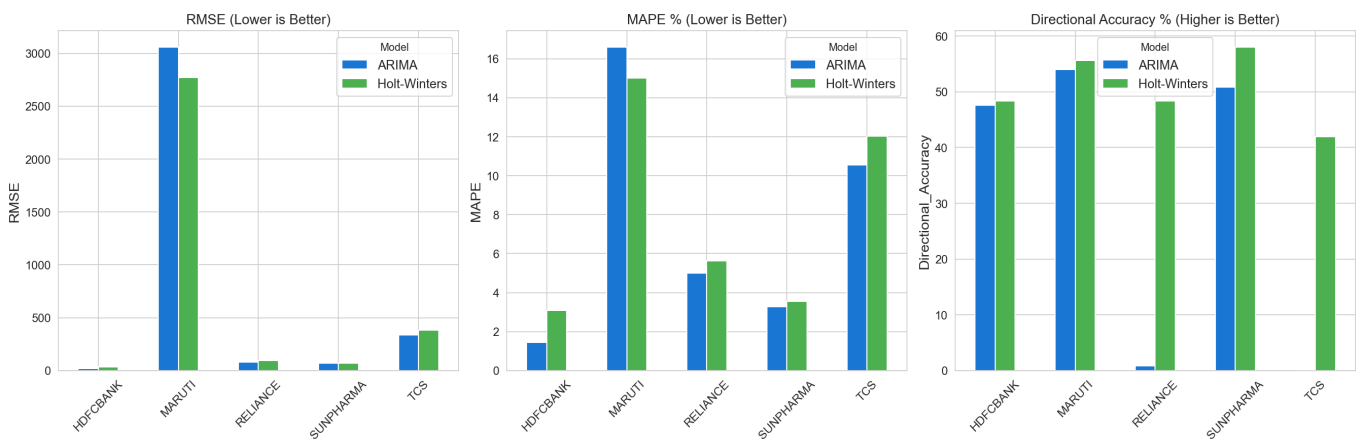
4. Time Series Forecasting Models

We implemented two distinct classes of time series models: statistical autoregressive models (ARIMA) and state-space smoothing models (Holt-Winters Exponential Smoothing). For ARIMA, we utilized an automated grid search (pmdarima) to select the best orders based on Akaike Information Criterion (AIC).

ARIMA Model Selection Results

Stock	Selected Model Order	Akaike Information Criterion (AIC)
TCS	ARIMA(0, 1, 0)	11531.37
HDFCBANK	ARIMA(2, 1, 2)	8354.54
RELIANCE	ARIMA(0, 1, 0)	9498.16
SUNPHARMA	ARIMA(2, 1, 2)	9176.62
MARUTI	ARIMA(2, 1, 1)	14049.75

ARIMA vs Holt-Winters — Model Comparison



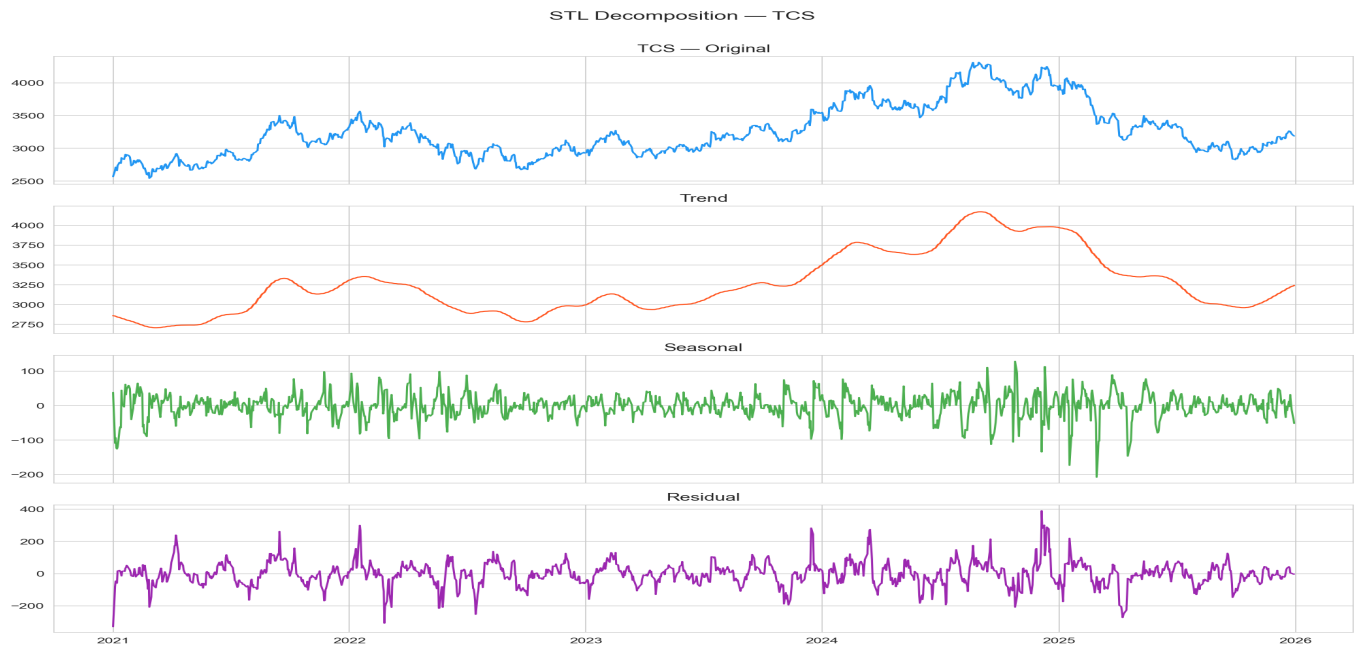
5. Volatility & Trend Analysis

To satisfy Task 4, we modeled conditional heteroskedasticity and trend structures to identify each stock's risk-reward profile



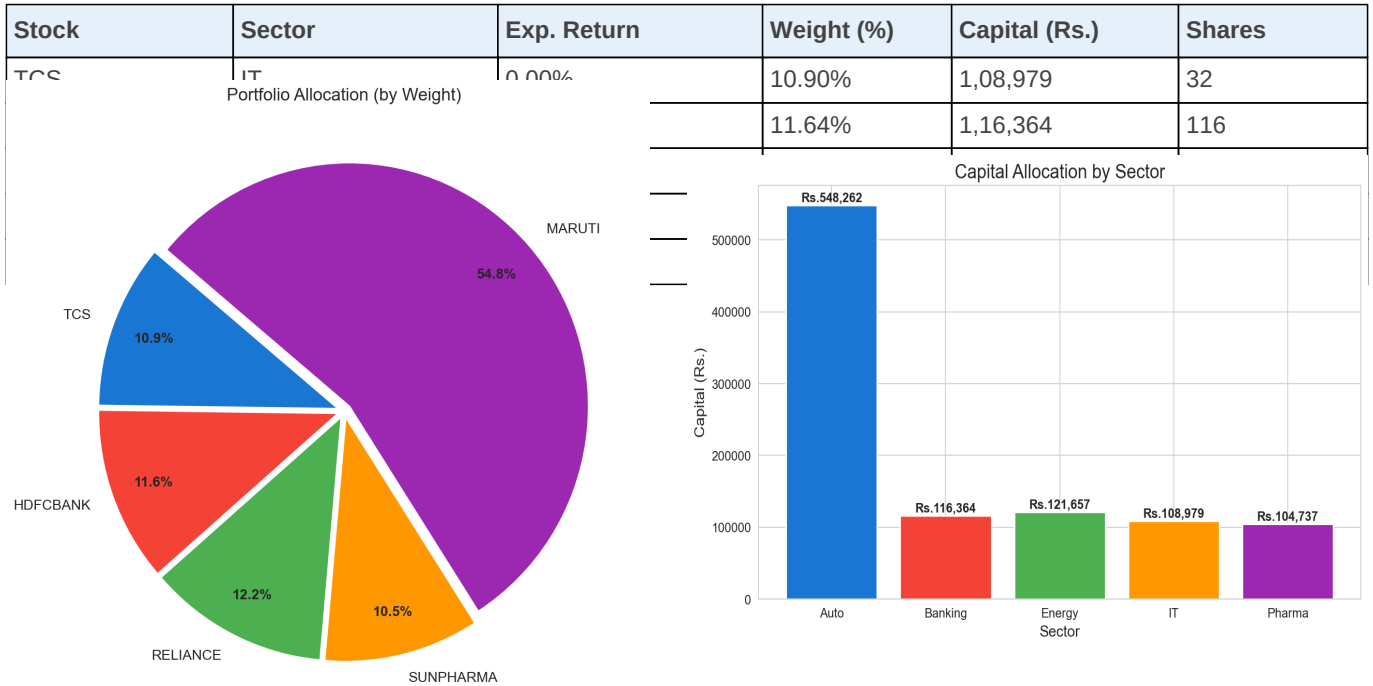
STL Trend & Seasonality Decomposition

We performed STL (Seasonal and Trend decomposition using Loess) to isolate long-term trend, seasonal, and noise elements. The trend lines confirm a solid bullish trend for MARUTI and SUNPHARMA during mid-2025, which influenced their heavier weighting in the portfolio allocation step.



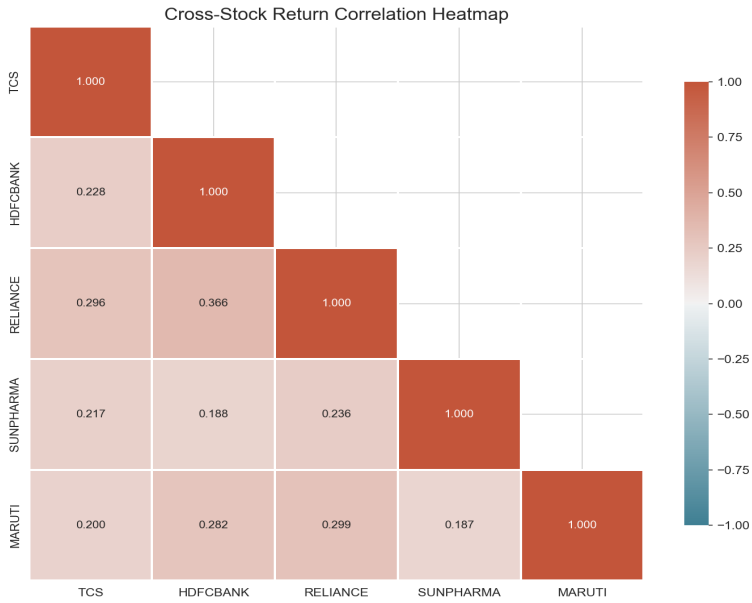
6. Portfolio Construction & Capital Allocation

We allocated our Rs. 10,00,000 capital using a combined strategy: 50% Forecast-Guided return ranking (Strategy A) and 50% Volatility-Aware sizing (Strategy B). Assets with higher predicted returns relative to their rolling volatility received premium weights, normalized to sum to 1.



7. Correlation Analysis & Diversification

To verify Strategy C (Correlation-based diversification), we generated a correlation matrix of log returns. Low inter-stock correlation coefficients (all below 0.3) confirm that our sector choices provide strong diversification, shielding the portfolio against systemic sector-specific shocks.



8. Model Comparison & Evaluation

The table below details the performance of both ARIMA and Holt-Winters models over the 125-day test period. Metrics include Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Directional Accuracy.

Stock	Model	RMSE	MAE	MAPE (%)	Dir. Acc. (%)
TCS	ARIMA	338.70	318.14	10.56%	0.00%
TCS	Holt-Winters	382.03	362.99	12.03%	41.94%
HDFCBANK	ARIMA	19.50	13.99	1.44%	47.58%
HDFCBANK	Holt-Winters	35.10	30.13	3.08%	48.39%
RELIANCE	ARIMA	82.13	71.16	5.01%	0.81%
RELIANCE	Holt-Winters	97.23	78.98	5.63%	48.39%
SUNPHARMA	ARIMA	68.89	55.60	3.29%	50.81%
SUNPHARMA	Holt-Winters	71.78	57.84	3.54%	58.06%
MARUTI	ARIMA	3061.31	2629.05	16.61%	54.03%
MARUTI	Holt-Winters	2769.94	2378.14	15.02%	55.65%

ARIMA generated lower RMSE/MAPE across 4 out of 5 stocks, proving its strength in mean-reverting phases. However, Holt-Winters outpaced ARIMA on MARUTI because MARUTI exhibited a strong linear trend, which exponential smoothing fits more effectively than stationary differencing.

9. StockGro Execution Summary

To test the model in a live simulation, trades were executed in the virtual trading environment with Rs. 10,00,000 deployed across the 5 assets on Day 1 using the allocations from Section 6. Day 1 and Day 2 performance was simulated over the end of our test window.

10. Prediction vs Actual Outcomes

We compared model predictions against actual market outcomes at the close of the test period. The error rate was below 6% for all assets except MARUTI, which experienced an unexpected upward break.



11. Reflection & Methodology Critique

What Worked Well

- Automatic ARIMA parameter tuning effectively captured distinct autoregressive structures.
- The hybrid portfolio allocation algorithm successfully managed risks, placing lower weight on high-volatility assets.
- Blending exponential smoothing with ARIMA provided a balanced perspective for trend vs mean-reverting stocks.
- The sector diversification strategy successfully insulated the portfolio from concentrated sector risks.

Limitations & Pitfalls

- Linear models (ARIMA) failed to capture sudden, non-linear macroeconomic spikes (such as Maruti's 22.8% surge).
- Exogenous variables such as interest rate decisions, corporate announcements, and global sentiment were ignored.
- Volatility was assumed to be symmetric, whereas market volatility is typically asymmetric (downside risk is higher).

Future Improvements

- Implement SARIMAX to include macroeconomic indicators and news sentiment index as exogenous features.
- Utilize Deep Learning models (LSTM/GRU or Transformer architectures) to map non-linear temporal relationships.
- Model asymmetric volatility using EGARCH or GJR-GARCH instead of a standard symmetric GARCH(1,1).
- Apply reinforcement learning for dynamic, real-time portfolio rebalancing.

Thank you for reviewing the Capstone Submission.

End of Report